



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta del 12/01/2022*

EXERCISE

We observe N samples of a realization of the random process $X[n] = s[n] + W[n]$, $n=0, 1, \dots, N-1$, with $N \gg 1$, where $s[n] = A \cos^2(2\pi F_0 n)$ is the signal of interest (SOI), A is the amplitude of the SOI and it is modelled as unknown deterministic parameters, and $W[n]$ is a zero-mean white Gaussian process with power σ_w^2 .

(1) Derive the **signal to noise ratio (SNR)** $\gamma \triangleq E\{\|\mathbf{s}\|^2\} / E\{\|\mathbf{W}\|^2\}$, where $\mathbf{s}$ and $\mathbf{W}$ are the vectors containing the N samples of signal and noise, respectively. Then, derive the **log-likelihood function** $LLF(A, \sigma_w^2) = \ln f_x(\mathbf{x}; A, \sigma_w^2)$, where $f_x(\mathbf{x}; A, \sigma_w^2)$ is the probability density function (PDF) of the observed data vector $\mathbf{X} = [X[0] \ X[1] \ \dots \ X[N-1]]^T$.

(2) Derive the **maximum likelihood (ML) estimator** of the signal amplitude A when the noise power σ_w^2 is a-priori known and provide a geometrical interpretation of it.

(3) Derive the **bias** and the **mean square error (MSE)** of the ML estimator of A ; establish if it is unbiased and consistent; plot the MSE of $\hat{A}_{ML}$ as a function of N and find the minimum value of N which guarantees an MSE lower than 10^{-2} when the noise power is $\sigma_w^2 = 5$.

(4) Assume now that also the noise power σ_w^2 is unknown: derive the **joint ML estimators** of A and σ_w^2 ; derive the bias of the ML estimator of σ_w^2 .

(5) Find the **Cramér-Rao bounds (CRBs)** on the joint estimation of A and σ_w^2 ; establish if the ML estimators of A is efficient in this case or not.

Useful relationships:

- 1) If $\mathbf{x}$ is a vector and $\mathbf{B}$ a square matrix: $\mathbf{x}^T \mathbf{B} \mathbf{x} = \text{Tr}\{\mathbf{B} \mathbf{x} \mathbf{x}^T\}$, $E\{\text{Tr}\{\mathbf{B}\}\} = \text{Tr}\{E\{\mathbf{B}\}\}$
- 2) If $\mathbf{P}$ is a projection matrix $\Rightarrow \text{Tr}\{\mathbf{P}\} = \text{rank}\{\mathbf{P}\}$



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Solution.

(1) Derive the SNR and the log-likelihood function.

$$s[n] = A \cos^2(2\pi F_0 n) \Rightarrow \mathbf{s} = A\mathbf{c}, \text{ where } [\mathbf{c}]_n = \cos^2(2\pi F_0 n) = \frac{1}{2} + \frac{1}{2} \cos(4\pi F_0 n)$$

$$E\{\|\mathbf{s}\|^2\} = \|\mathbf{s}\|^2 = A^2 \left(\frac{1}{4} \cdot N + \frac{1}{4} \cdot \frac{N}{2} \right) = \frac{3N}{8} A^2, \quad E\{\|\mathbf{W}\|^2\} = N\sigma_w^2, \quad \gamma \triangleq E\{\|\mathbf{s}\|^2\} / E\{\|\mathbf{W}\|^2\} = \frac{3A^2}{8\sigma_w^2}$$

$$X[n] \in N(s[n], \sigma_w^2), \quad \{X[n]\}_{n=0}^{N-1} \text{ independent} \Rightarrow \mathbf{X} \in N(A\mathbf{c}, \sigma_w^2 \mathbf{I})$$

PDF in scalar format:

$$f_X(\mathbf{x}; A, \sigma_w^2) = \prod_{n=0}^{N-1} f_X(x[n]; A, \sigma_w^2) = \left(\frac{1}{\sqrt{2\pi\sigma_w^2}} \right)^N \exp\left(-\frac{1}{2\sigma_w^2} \sum_{n=0}^{N-1} (x[n] - A \cos^2(2\pi F_0 n))^2 \right)$$

PDF in vector format:

$$f_X(\mathbf{x}; A, \sigma_w^2) = \frac{1}{(2\pi)^{N/2} \sqrt{|\sigma_w^2 \mathbf{I}|}} \exp\left(-\frac{1}{2} (\mathbf{x} - A\mathbf{c})^T (\sigma_w^2 \mathbf{I})^{-1} (\mathbf{x} - A\mathbf{c}) \right) = \frac{1}{(2\pi)^{N/2} (\sigma_w^2)^{N/2}} \exp\left(-\frac{1}{2\sigma_w^2} \|\mathbf{x} - A\mathbf{c}\|^2 \right)$$

The log-likelihood function is given by:

$$LLF(A, \sigma_w^2) = \ln f_X(\mathbf{x}; A, \sigma_w^2) = -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\sigma_w^2) - \frac{1}{2\sigma_w^2} \|\mathbf{x} - A\mathbf{c}\|^2$$

(2) Derive the maximum likelihood (ML) estimator of the signal amplitude A when the noise power σ_w^2 is a-priori known.

$$\hat{A}_{ML} = \hat{A}_{LS} = \mathbf{c}^H \mathbf{X} = (\mathbf{c}^T \mathbf{c})^{-1} \mathbf{c}^T \mathbf{X} = \frac{1}{\|\mathbf{c}\|^2} \mathbf{c}^T \mathbf{X} = \frac{8}{3N} \mathbf{c}^T \mathbf{X} = \frac{8}{3N} \sum_{n=0}^{N-1} x[n] \cos^2(2\pi F_0 n)$$

A is estimated at the output of a linear matched filter: it is given by the orthogonal projection of the received signal $\mathbf{X}$ onto the signal subspace, i.e. the subspace generated by vector $\mathbf{c}$. The estimator is linear, so it is Gaussian distributed.

(3) Derive the bias and the mean square error (MSE) of the ML estimator of A and establish if it is unbiased and consistent; plot the MSE of $\hat{A}_{ML}$ as a function of N and find the minimum value of N which guarantees an MSE lower than 10^{-2} when the noise power is $\sigma_w^2 = 5$.



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$$E\{\hat{A}_{ML}\} = E\left\{\frac{1}{\|\mathbf{c}\|^2} \mathbf{c}^T \mathbf{X}\right\} = \frac{1}{\|\mathbf{c}\|^2} \mathbf{c}^T E\{\mathbf{X}\} = \frac{1}{\|\mathbf{c}\|^2} \mathbf{c}^T (A\mathbf{c}) = A$$

The estimator is unbiased.

$$\begin{aligned} MSE\{\hat{A}_{ML}\} &= E\left\{\left(A - \hat{A}_{ML}\right)^2\right\} = E\left\{\left(A - \frac{\mathbf{c}^T \mathbf{X}}{\|\mathbf{c}\|^2}\right)^2\right\} = E\left\{\left(A - \frac{\mathbf{c}^T}{\|\mathbf{c}\|^2} (A\mathbf{c} + \mathbf{W})\right)^2\right\} = E\left\{\left(A - A - \frac{\mathbf{c}^T \mathbf{W}}{\|\mathbf{c}\|^2}\right)^2\right\} \\ &= E\left\{\left(-\frac{\mathbf{c}^T \mathbf{W}}{\|\mathbf{c}\|^2}\right)^2\right\} = \frac{1}{\|\mathbf{c}\|^4} E\left\{\mathbf{c}^T \mathbf{W} (\mathbf{c}^T \mathbf{W})^T\right\} = \frac{1}{\|\mathbf{c}\|^4} E\left\{\mathbf{c}^T \mathbf{W} \mathbf{W}^T \mathbf{c}\right\} = \frac{\sigma_w^2}{\|\mathbf{c}\|^4} \mathbf{c}^T \mathbf{c} = \frac{\sigma_w^2}{\|\mathbf{c}\|^2} = \frac{8\sigma_w^2}{3N} \end{aligned}$$

The estimator is consistent.

$$\sigma_w^2 = 5: \quad MSE\{\hat{A}_{ML}\} = \frac{40}{3N} \leq 10^{-2} \quad \Rightarrow \quad N \geq \frac{4000}{3} = 1334$$

(4) Assume now that also the noise power σ_w^2 is unknown: derive the joint ML estimator of A and σ_w^2 ; derive the bias of the ML estimator of σ_w^2 .

$$LLF(A, \sigma_w^2) = \ln f_X(\mathbf{x}; A, \sigma_w^2) = -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\sigma_w^2) - \frac{1}{2\sigma_w^2} \|\mathbf{x} - A\mathbf{c}\|^2$$

$$\frac{dLLF(A, \sigma_w^2)}{d\sigma_w^2} = -\frac{N}{2\sigma_w^2} + \frac{1}{2\sigma_w^4} \|\mathbf{x} - A\mathbf{c}\|^2 = 0 \quad \Rightarrow \quad \sigma_w^2 = \frac{1}{N} \|\mathbf{x} - A\mathbf{c}\|^2$$

Replacing A with its ML estimate we get the ML estimate of the noise power:

$$\hat{\sigma}_{w,ML}^2 = \frac{1}{N} \|\mathbf{x} - \hat{A}_{ML} \mathbf{c}\|^2 = \frac{1}{N} \left\| \mathbf{x} - \frac{\mathbf{c} \mathbf{c}^T \mathbf{x}}{\|\mathbf{c}\|^2} \right\|^2 = \frac{1}{N} \|\mathbf{x} - \mathbf{P}_c \mathbf{x}\|^2 = \frac{1}{N} \|(\mathbf{I} - \mathbf{P}_c) \mathbf{x}\|^2 = \frac{1}{N} \mathbf{x}^T \mathbf{P}_c^\perp \mathbf{x}$$

where $\mathbf{P}_c^\perp = \mathbf{I} - \mathbf{P}_c = \mathbf{I} - \frac{\mathbf{c} \mathbf{c}^T}{\|\mathbf{c}\|^2} = \mathbf{I} - \mathbf{c} \mathbf{c}^\#$ is the projection matrix onto the $(N-1)$ -dimensional noise

subspace, i.e. the subspace orthogonal to the 1-D signal subspace generated by vector $\mathbf{c}$. The ML estimator of σ_w^2 projects the received data vector $\mathbf{X}$ onto the noise subspace and then calculates the energy of the data in the noise subspace, i.e. the sum of the powers of the $N-1$ components where only the noise is present; finally the power is obtained by dividing the energy by the length of the observation time N . Since the signal subspace is 1-D, the orthogonal noise subspace is $(N-1)$ -D, hence the bias of the ML estimator can be found as follows:



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$$\begin{aligned} E\{\mathbf{X}^T \mathbf{P}_c^\perp \mathbf{X}\} &= E\{Tr\{\mathbf{P}_c^\perp \mathbf{X} \mathbf{X}^T\}\} = Tr\{E\{\mathbf{P}_c^\perp \mathbf{X} \mathbf{X}^T\}\} = Tr\{\mathbf{P}_c^\perp E\{\mathbf{X} \mathbf{X}^T\}\} \\ &= Tr\{\mathbf{P}_c^\perp (A^2 \mathbf{c} \mathbf{c}^T + \sigma_w^2 \mathbf{I})\} = Tr\{\mathbf{P}_c^\perp (\sigma_w^2 \mathbf{I})\} = \sigma_w^2 Tr\{\mathbf{P}_c^\perp\} = \sigma_w^2 (N-1) \end{aligned}$$

Note that $\mathbf{P}_c^\perp \mathbf{c} = 0$, since $\mathbf{P}_c^\perp$ is the projection matrix onto the $(N-1)$ -dimensional noise subspace, i.e. the subspace orthogonal to vector $\mathbf{c}$. Hence, the bias is immediately derived as follows:

$$E\{\hat{\sigma}_{w,ML}^2\} = E\left\{\frac{1}{N} \mathbf{X}^T \mathbf{P}_c^\perp \mathbf{X}\right\} = \frac{\sigma_w^2 (N-1)}{N} \Rightarrow bias\{\hat{\sigma}_{w,ML}^2\} = \sigma_w^2 - \frac{\sigma_w^2 (N-1)}{N} = \frac{\sigma_w^2}{N}$$

(5) Find the Cramér-Rao bounds (CRBs) on the joint estimation of A and σ_w^2 and establish if the ML estimators of A is efficient or not.

$$\begin{aligned} LLF(A, \sigma_w^2) &= \ln f_{\mathbf{x}}(\mathbf{x}; A, \sigma_w^2) = -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\sigma_w^2) - \frac{1}{2\sigma_w^2} \|\mathbf{x} - A\mathbf{c}\|^2 \\ &= -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\sigma_w^2) - \frac{1}{2\sigma_w^2} (\mathbf{x} - A\mathbf{c})^T (\mathbf{x} - A\mathbf{c}) \\ &= -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\sigma_w^2) - \frac{1}{2\sigma_w^2} (\mathbf{x}^T \mathbf{x} - 2A\mathbf{c}^T \mathbf{x} + A^2 \mathbf{c}^T \mathbf{c}) \end{aligned}$$

$$\frac{dLLF(A, \sigma_w^2)}{dA} = -\frac{1}{2\sigma_w^2} (-2\mathbf{c}^T \mathbf{x} + 2A\mathbf{c}^T \mathbf{c}), \quad \frac{d^2 LLF(A, \sigma_w^2)}{dA^2} = -\frac{\mathbf{c}^T \mathbf{c}}{\sigma_w^2} = -\frac{3N}{8\sigma_w^2}$$

$$\frac{dLLF(A, \sigma_w^2)}{d\sigma_w^2} = -\frac{N}{2\sigma_w^2} + \frac{1}{2\sigma_w^4} \|\mathbf{x} - A\mathbf{c}\|^2, \quad \frac{d^2 LLF(A, \sigma_w^2)}{(d\sigma_w^2)^2} = \frac{N}{2\sigma_w^4} - \frac{1}{\sigma_w^6} \|\mathbf{x} - A\mathbf{c}\|^2$$

$$\frac{d^2 LLF(A, \sigma_w^2)}{dAd\sigma_w^2} = \frac{d}{d\sigma_w^2} \left(-\frac{1}{2\sigma_w^2} (-2\mathbf{c}^T \mathbf{x} + 2A\mathbf{c}^T \mathbf{c}) \right) = \frac{1}{2\sigma_w^4} (-2\mathbf{c}^T \mathbf{x} + 2A\mathbf{c}^T \mathbf{c})$$

$$I_{11} = -E\left\{\frac{d^2 LLF(A, \sigma_w^2)}{dA^2}\right\} = \frac{3N}{8\sigma_w^2}$$

$$I_{22} = -E\left\{\frac{d^2 LLF(A, \sigma_w^2)}{(d\sigma_w^2)^2}\right\} = -\frac{N}{2\sigma_w^4} + \frac{1}{\sigma_w^6} E\{\|\mathbf{x} - A\mathbf{c}\|^2\} = -\frac{N}{2\sigma_w^4} + \frac{1}{\sigma_w^6} E\{\|\mathbf{w}\|^2\} = -\frac{N}{2\sigma_w^4} + \frac{N\sigma_w^2}{\sigma_w^6} = \frac{N}{2\sigma_w^4}$$

$$I_{12} = I_{21} = -E\left\{\frac{d^2 LLF(A, \sigma_w^2)}{dAd\sigma_w^2}\right\} = -\frac{1}{2\sigma_w^4} (-2\mathbf{c}^T E\{\mathbf{x}\} + 2A\mathbf{c}^T \mathbf{c}) - \frac{1}{2\sigma_w^4} (-2A\mathbf{c}^T \mathbf{c} + 2A\mathbf{c}^T \mathbf{c}) = 0$$

Since $I_{12} = 0$, the two parameters are decoupled and the two CRB are:

$$CRB(A) = \frac{8\sigma_w^2}{3N}, \quad CRB(\sigma_w^2) = \frac{2\sigma_w^4}{N}. \quad \text{Hence, the ML estimator of } A \text{ is efficient.}$$